

Lab_W2_Designing_MapReduce_Algorithms_SGonzales

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1 IMT 575 Lab 2.17 - Steve Gonzales

1.1 Introduction

In this lab, you will design and implement MapReduce algorithms for a variety of common data processing tasks.

The MapReduce programming model (and a corresponding system) was proposed in a 2004 paper from a team at Google as a simpler abstraction for processing very large datasets in parallel. The goal of this assignment is to give you experience “thinking in MapReduce.” We will use small datasets that you can inspect directly to determine the correctness of your results and to internalize how MapReduce works.

You may want to review the material in module 2.16 to make sure you understand the programming model.

1.2 Setup

Download the data.zip file we will use in this assignment from Canvas.

To keep things simple, place all files in the same directory as this notebook.

1.3 A Simplified MapReduce Framework

Our goal is to practice thinking in terms of mapreduce without grappling with the systems issues associated with installing and configuring a complete distributed MapReduce environment.

To that end, we’re providing you a simple Python MapReduce framework that returns results like the real thing, but does not involve parallel processing. We’re writing programs that could be parallelized using MapReduce, but instead we’re going to run them locally in Python.

MapReduce is a framework, not a programming language. The concepts can be used in any programming language. We’re using Python here for convenience.

The next cell provides a class definition that implements a simple MapReduce framework.

```
[1]: import json
```

```
[2]: list_ex = []  
     type(list_ex)  
     tuple_ex = ()  
     type(tuple_ex)
```

```

print("List object implements: ", ', '.join([m for m in dir(list_ex) if not m.
↳startswith('_')]))
print("Tuple object implements: ", ', '.join([m for m in dir(tuple_ex) if not m.
↳startswith('_')]))

dict_ex = {}
type(dict_ex)
print("Dictionary object implements: ", ', '.join([m for m in dir(dict_ex) if
↳not m.startswith('_')]))
dict_ex['a'] = ['abc']
dict_ex
# set_ex = {1, 3, 'a'}
# type(set_ex)

```

List object implements: append, clear, copy, count, extend, index, insert, pop, remove, reverse, sort

Tuple object implements: count, index

Dictionary object implements: clear, copy, fromkeys, get, items, keys, pop, popitem, setdefault, update, values

[2]: {'a': ['abc']}

```

[3]: # Main class for MapReduce functionality
class MapReduce:
    def __init__(self):
        self.intermediate = {}
        self.result = []

    def emit_intermediate(self, key, value):
        self.intermediate.setdefault(key, [])
        self.intermediate[key].append(value)

    def emit(self, value):
        self.result.append(value)

    def execute(self, data, mapper, reducer):
        result = []
        for line in data:
            record = json.loads(line)
            mapper(record)

        for key in self.intermediate:
            reducer(key, self.intermediate[key])

        #jenc = json.JSONEncoder(encoding='latin-1')
        jenc = json.JSONEncoder()

```

```

    for item in self.result:
        print(jenc.encode(item))
        # Added result output to use to compare to known answers
        result.append(json.loads(jenc.encode(item)))
    return result

```

1.4 Helper Functions

```

[4]: # The answer keys are large and it's impossible to tell if you have a correct
    ↪ output unless one does a row by row comparison.
    # This function compares two lists of lists and returns a dictionary with the
    ↪ differences.
def compare_list(list1, list2):
    """Compare two multidimensional lists and return True if they are equal,
    ↪ False otherwise.
    The lists could be out of order. Display only the lines that are different.
    """
    # Create a dict from the second list for quick lookup
    d = {}
    d['1'] = {item[0]: item[1] for item in list1}
    d['2'] = {item[0]: item[1] for item in list2}
    result = {'not_in_list1': [], 'not_in_list2': [], 'different_keys': []}
    for d_name in ['1', '2']:
        other_d_name = '1' if d_name == '2' else '1'
        for k, v in d[d_name].items():
            if k not in d[other_d_name]:
                print(f"Key {k} not found in list{other_d_name}.")
                result[f'not_in_list{other_d_name}'].append(k)
            elif sorted(d[other_d_name][k]) != sorted(v):
                print(f"Different values for key {k}: {v} vs
    ↪ {d[other_d_name][k]}")
                result['different_keys'].append(k)
    result = {'not_in_list1': sorted(result['not_in_list1']), 'not_in_list2':
    ↪ sorted(result['not_in_list2']), 'different_keys':
    ↪ sorted(result['different_keys'])}
    return result

# The files are not JSON, but are a list of lists, but not even enclosed in
    ↪ brackets nor do they have commas separating elements.
# This function converts the hacked files to a Python list.
def convert_hack_to_list(hacked_file):
    """Convert a hacked file to a list of dictionaries."""
    result = []
    with open(hacked_file, 'r', encoding='utf-8') as f:
        for line in f:
            line = line.strip()

```

```

        if not line:
            continue
        result.append(eval(line))
    return result

# Mini Unit Test
answer_file = 'inverted_index.json'
answers = convert_hack_to_list(answer_file)[:3]
print(answers[0], type(answers[0]))
attempt = [["Rossmore", ["edgeworth-parents.txt"]]]
r = compare_list(attempt, answers)
print(r)

['all', ['milton-paradise.txt', 'blake-poems.txt', 'melville-moby_dick.txt']]
<class 'list'>
Key all not found in list1.
Key Consumptive not found in list1.
{'not_in_list1': ['Consumptive', 'all'], 'not_in_list2': [], 'different_keys':
[]}
```

1.5 Example: Word Count

The next cell contains an example of using the class above.

In Part 0, set up the name of the file. Modify this if you downloaded the assignment data to a different location.

In Part 1, we create a MapReduce object that is used to pass data between the map function and the reduce function; you won't need to use this object directly.

In Part 2, the mapper function tokenizes each document and emits a key-value pair. The key is a word formatted as a string and the value is the integer 1 to indicate an occurrence of word.

In Part 3, the reducer function sums up the list of occurrence counts and emits a count for word. Since the mapper function emits the integer 1 for each word, each element in the list_of_values is the integer 1.

The list of occurrence counts is summed and a (word, total) tuple is emitted where word is a string and total is an integer.

When you run this cell, you should see the results printed to the screen: a sequence of pairs (word, count) for every word found in the dataset.

```

[5]: # Part 0
filename = "books.json" # modify this if needed

# Part 1
mr = MapReduce()

# Part 2
def mapper(record):
```

```

    # key: document identifier
    # value: document contents
    key = record[0]
    value = record[1]
    words = value.split()
    for w in words:
        mr.emit_intermediate(w, 1)

# Part 3
def reducer(key, list_of_values):
    # key: word
    # value: list of occurrence counts
    total = 0
    for v in list_of_values:
        total += v
    mr.emit((key, total))

# Part 4
with open(filename) as inputdata:
    result = mr.execute(inputdata, mapper, reducer)
print(len(result))

```

```

["", 11]
["Paradise", 1]
["Lost", 1]
["by", 12]
["John", 1]
["Milton", 1]
["1667", 1]
["", 11]
["Book", 2]
["I", 25]
["Of", 4]
["Man", 3]
["'", 12]
["s", 7]
["first", 4]
["disobedience", 1]
["", 101]
["and", 38]
["the", 58]
["fruit", 1]
["that", 13]
["forbidden", 1]
["tree", 1]
["whose", 1]
["mortal", 1]
["taste", 1]

```

["Brought", 1]
["death", 2]
["into", 1]
["World", 1]
["all", 4]
["our", 2]
["woe", 1]
["With", 1]
["loss", 1]
["of", 35]
["Eden", 1]
["till", 1]
["one", 3]
["greater", 1]
["Restore", 1]
["us", 3]
["regain", 1]
["blissful", 1]
["seat", 1]
["Sing", 3]
["Heavenly", 1]
["Muse", 1]
["on", 4]
["secret", 1]
["top", 1]
["Oreb", 1]
["or", 4]
["Sinai", 1]
["didst", 1]
["inspire", 1]
["That", 4]
["shepherd", 1]
["who", 2]
["taught", 1]
["chosen", 1]
["seed", 1]
["In", 3]
["beginning", 2]
["how", 1]
["heavens", 1]
["earth", 4]
["Rose", 1]
["out", 1]
["Chaos", 1]
[":", 14]
["if", 1]
["Sion", 1]
["hill", 1]

["Delight", 1]
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["more", 2]
["Siloa", 1]
["brook", 1]
["flowed", 1]
["Fast", 1]
["oracle", 1]
["God", 7]
["thence", 1]
["Invoke", 1]
["thy", 7]
["aid", 1]
["to", 20]
["my", 6]
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["no", 2]
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["while", 1]
["it", 10]
["pursues", 1]
["Things", 1]
["unattempted", 1]
["yet", 2]
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["prose", 1]
["rhyme", 1]
[".", 46]
["The", 13]
["Parent", 1]
["Assistant", 1]
["Maria", 1]
["Edgeworth", 1]
["THE", 3]
["ORPHANS", 1]
["Near", 1]
["ruins", 1]
["castle", 1]
["Rossmore", 1]

["Ireland", 1]
["is", 3]
["a", 29]
["small", 1]
["cabin", 1]
["which", 2]
["there", 3]
["once", 1]
["lived", 2]
["widow", 1]
["her", 8]
["four", 1]
["children", 1]
["As", 1]
["long", 4]
["as", 7]
["she", 5]
["was", 16]
["able", 1]
["work", 1]
["very", 4]
["industrious", 1]
["accounted", 1]
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["parish", 1]
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["Mary", 1]
["Emma", 2]

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["Austen", 1]
["1816", 1]
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["CHAPTER", 1]
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["rich", 1]
["comfortable", 1]
["home", 3]
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["existence", 1]
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["-", 4]
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["vex", 1]
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["indulgent", 1]
["father", 1]
["consequence", 1]
["sister", 1]
["marriage", 1]
["been", 2]
["mistress", 1]
["his", 4]
["house", 1]
["from", 3]
["early", 1]
["period", 1]
["Her", 1]
["mother", 2]
["died", 1]

["too", 1]
["ago", 1]
["for", 5]
["have", 1]
["than", 1]
["an", 2]
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["remembrance", 1]
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["fallen", 1]
["short", 1]
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["K", 2]
["Chesterton", 2]
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["IN", 1]
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["Lucifer", 1]
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["James", 2]
["Bible", 2]
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["Testament", 1]
["First", 1]
["Moses", 1]
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["Genesis", 1]
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["upon", 3]
["face", 2]
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["joy", 2]
["write", 2]
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[".\\"", 1]
["vanish", 1]
["d", 4]
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["pluck", 1]
["hollow", 1]
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["water", 1]
["clear", 1]
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["Caesar", 1]
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["Scoena", 1]
["Prima", 1]
["Enter", 1]
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["Murellus", 1]
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["Commoners", 1]
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["Cobler", 1]
["But", 1]
["Answer", 1]

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["Whitman", 2]
["1855", 1]
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["winds", 1]
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["waves", 1]
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["now", 2]

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["En", 1]
["Masse", 1]
["Moby", 1]
["Dick", 1]
["Herman", 1]
["Melville", 1]
["1851", 1]
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["Late", 1]
["Consumptive", 1]
["Usher", 2]
["Grammar", 1]
["School", 1]
["pale", 1]
["threadbare", 1]
["coat", 1]
["heart", 1]
["body", 1]
["brain", 1]
["see", 1]
["him", 2]
["He", 2]
["dusting", 1]
["lexicons", 1]
["grammars", 2]
["queer", 1]
["handkerchief", 1]
["mockingly", 1]
["embellished", 1]

```
["flags", 1]
["known", 1]
["nations", 1]
["loved", 1]
["dust", 1]
["somehow", 1]
["mildly", 1]
["reminded", 1]
["mortality", 1]
611
```

1.6 Problem 1: Inverted Index

(6 points)

Create an inverted index. Given a set of documents, an inverted index is a dictionary where each word is associated with a list of the document identifiers in which that word appears.

1.6.1 Mapper Input

The input is a two-element list: [document_id, text], where document_id is a string representing a document identifier and text is a string representing the text of the document. The document text may have words in upper or lower case and may contain punctuation. You should treat each token as if it was a valid word; that is, you can just use `value.split()` to tokenize the string.

1.6.2 Reducer Output

The output should be a (word, document ID list) tuple where word is a String and document ID list is a list of Strings.

The correct output can be found in `inverted_index.json`

```
[102]: # Part 0
filename = "books.json"

# Part 1
mr = MapReduce()

# Part 2
def mapper(record):
    # key: document identifier
    # value: document contents
    doc_id = record[0]
    text = record[1]
    for word in text.split():
        mr.emit_intermediate(word, doc_id)

# Part 3
def reducer(key, list_of_values):
    # key: word
```

```

# value: list of document ids containing word
unique_doc_ids = sorted(set(list_of_values))
mr.emit((key, unique_doc_ids))

# Part 4
with open(filename) as inputdata:
    invert_res = mr.execute(inputdata, mapper, reducer)
# Check against answer file
answers = convert_hack_to_list('inverted_index.json')
print(compare_list(invert_res, answers))

[["", ["austen-emma.txt", "bible-kjv.txt", "blake-poems.txt", "chesterton-
ball.txt", "chesterton-thursday.txt", "edgeworth-parents.txt", "melville-
moby_dick.txt", "milton-paradise.txt", "shakespeare-caesar.txt", "whitman-
leaves.txt"]],
["Paradise", ["milton-paradise.txt"]],
["Lost", ["milton-paradise.txt"]],
["by", ["austen-emma.txt", "blake-poems.txt", "chesterton-ball.txt",
"chesterton-thursday.txt", "edgeworth-parents.txt", "melville-moby_dick.txt",
"milton-paradise.txt", "shakespeare-caesar.txt", "whitman-leaves.txt"]],
["John", ["milton-paradise.txt"]],
["Milton", ["milton-paradise.txt"]],
["1667", ["milton-paradise.txt"]],
["", ["austen-emma.txt", "bible-kjv.txt", "blake-poems.txt", "chesterton-
ball.txt", "chesterton-thursday.txt", "edgeworth-parents.txt", "melville-
moby_dick.txt", "milton-paradise.txt", "shakespeare-caesar.txt", "whitman-
leaves.txt"]],
["Book", ["bible-kjv.txt", "milton-paradise.txt"]],
["I", ["austen-emma.txt", "blake-poems.txt", "chesterton-ball.txt", "chesterton-
thursday.txt", "melville-moby_dick.txt", "milton-paradise.txt", "shakespeare-
caesar.txt", "whitman-leaves.txt"]],
["Of", ["milton-paradise.txt", "shakespeare-caesar.txt"]],
["Man", ["chesterton-thursday.txt", "milton-paradise.txt"]],
["'", ["austen-emma.txt", "blake-poems.txt", "edgeworth-parents.txt", "milton-
paradise.txt", "whitman-leaves.txt"]],
["s", ["austen-emma.txt", "edgeworth-parents.txt", "milton-paradise.txt",
"whitman-leaves.txt"]],
["first", ["bible-kjv.txt", "milton-paradise.txt", "whitman-leaves.txt"]],
["disobedience", ["milton-paradise.txt"]],
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["Drop", ["blake-poems.txt"]]
[":!\\", ["blake-poems.txt"]]
["same", ["blake-poems.txt"]]
["While", ["blake-poems.txt"]]
["joy", ["blake-poems.txt"]]

```

```

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["read", ["blake-poems.txt"]]
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["vanish", ["blake-poems.txt"]]
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["ouer", ["shakespeare-caesar.txt"]]
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["Hence", ["shakespeare-caesar.txt"]]
["idle", ["shakespeare-caesar.txt"]]
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["What", ["shakespeare-caesar.txt"]]
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```

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 ["Or", ["whitman-leaves.txt"]]
 ["hence", ["whitman-leaves.txt"]]
 ["other", ["whitman-leaves.txt"]]
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 ["waves", ["whitman-leaves.txt"]]
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["coat", ["melville-moby_dick.txt"]]
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["He", ["melville-moby_dick.txt"]]
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["mildly", ["melville-moby_dick.txt"]]
["reminded", ["melville-moby_dick.txt"]]
["mortality", ["melville-moby_dick.txt"]]
{'not_in_list1': [], 'not_in_list2': [], 'different_keys': []}

```


1.7 Problem 2: Relational Join

(6 points)

Implement a relational join as a MapReduce query.

Consider the following query: `SELECT * FROM Orders, LineItem WHERE Order.order_id = LineItem.order_id`

This query produces all pairs of records from a table `Orders` and a table `LineItem` such that the `order_id` of the order matches the `order_id` of the `LineItem`. This *join* operation is ubiquitous in databases.

Your MapReduce query should produce a result similar to this SQL query would return when executed against an appropriate database.

You can consider the two input tables, `Order` and `LineItem`, as one big concatenated bag of records that will be processed by the Map function record by record. That is, instead of two files, each with a separate table, you are given one big file with both tables inside it. Each record is labeled with which table it came from. For example, here are two lines in `records.json`:

```
[ "order", "2", "78002", "0", "46929.18", "1996-12-01", "1-URGENT", "Clerk#000000880", "0", " f  
[ "line_item", "2", "106170", "1191", "1", "38", "44694.46", "0.00", "0.05", "N", "0", "1997-01-
```

The first element of each record indicates which table the record is stored in. The other elements are the attribute values for this record. This example shows a key difference between databases and MapReduce: there is no schema in most MapReduce systems! You can put in any kind of crazy data you wish, but the tradeoff is that the database can't help protect you from bad data – without a schema, it can't tell what's good and what's bad.

1.7.1 Map Input

Each input record is a list of strings representing a tuple in the database. Each list element corresponds to a different attribute of the table

The first item (index 0) in each record is a string that identifies the table the record originates from. This field has two possible values:

“line_item” indicates that the record is a line item. “order” indicates that the record is an order.

The second element (index 1) in each record is the `order_id`.

`LineItem` records have 17 attributes including the identifier string.

`Order` records have 10 elements including the identifier string.

1.7.2 Reduce Output

The output should be a joined record: a single list of length 27 that contains the attributes from the order record followed by the fields from the line item record. Each list element should be a string.

The correct output can be found in `join.json`

```
[103]: # Part 0
filename = "records.json"

# Part 1
mr = MapReduce()

# Part 2
def mapper(record):
    # key: None
    # value: a complete record, either an order or a line item
    mr.emit_intermediate(record[1], record)

# Part 3
def reducer(key, list_of_values):
    orders = [r for r in list_of_values if r[0] == "order"]
    line_items = [r for r in list_of_values if r[0] == "line_item"]
    for order in orders:
        for line_item in line_items:
            # Exclude the table name from both records
            joined = order + line_item
            mr.emit((joined))

# Part 4
with open(filename) as inputdata:
    join_list = mr.execute(inputdata, mapper, reducer)
# Check against answer file
answers = convert_hack_to_list('join.json')
for row in join_list:
    found = False
    for answer in answers:
        if row == answer:
            found = True
            break
    if not found:
        print(f"**Row {row} not found in answers.")
```

```
["order", "1", "36901", "0", "173665.47", "1996-01-02", "5-LOW",
"Clerk#000000951", "0", "nstructions sleep furiously among ", "line_item", "1",
"155190", "7706", "1", "17", "21168.23", "0.04", "0.02", "N", "0", "1996-03-13",
"1996-02-12", "1996-03-22", "DELIVER IN PERSON", "TRUCK", "egular courts above
the"]
["order", "1", "36901", "0", "173665.47", "1996-01-02", "5-LOW",
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"67310", "7311", "2", "36", "45983.16", "0.09", "0.06", "N", "0", "1996-04-12",
"1996-02-28", "1996-04-20", "TAKE BACK RETURN", "MAIL", "ly final dependencies:
slyly bold "]
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```

"63700", "3701", "3", "8", "13309.60", "0.10", "0.02", "N", "0", "1996-01-29",
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 express dep"]
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gifts cajole carefully."]

```

1.8 Problem 3: Counting Friends

(6 points)

Consider a simple social network dataset consisting of a set of key-value pairs (person, friend) representing a friend relationship between two people. Describe a MapReduce algorithm to count the number of friends for each person.

1.8.1 Map Input

Each input record is a pair (personA, personB) where personA is a string representing the name of a person and personB is a string representing the name of one of personA's friends. Note that it may or may not be the case that the personA is a friend of personB.

1.8.2 Reduce Output

The output should be a pair (person, friend_count) where person is a string and friend_count is an integer indicating the number of friends associated with person.

The correct output can be found in friend_count.json

```
[104]: # Part 0
filename = "friends.json"

# Part 1
mr = MapReduce()

# Part 2
def mapper(record):
    # key: personA
    # value: personB
    mr.emit_intermediate(record[0], record[1])

# Part 3
def reducer(key, list_of_values):
    mr.emit((key, len(list_of_values)))

# Part 4
with open(filename) as inputdata:
    friend_list = mr.execute(inputdata, mapper, reducer)
# Check against answer file
answers = convert_hack_to_list('friend_count.json')
for row in friend_list:
    found = False
    for answer in answers:
        if row == answer:
            found = True
            break
    if not found:
        print(f"**Row {row} not found in answers.")
```

```
["Myriel", 5]
["Napoleon", 1]
["MlleBaptistine", 3]
["MmeMagloire", 1]
["Champtercier", 1]
["Valjean", 16]
```

```
[105]: # Problem 3 but count unique friends with a modified file with duplicates
filename = "friends_dupe.json"

def reducer_unique(key, list_of_values):
    unique_friends = set(list_of_values)
    mr.emit((key, len(unique_friends)))

print("Counting non unique friends...")
mr = MapReduce()
with open(filename) as inputdata:
    friend_list = mr.execute(inputdata, mapper, reducer)
print("")

print("Counting unique friends...")
mr = MapReduce()
with open(filename) as inputdata:
    unique_friend_list = mr.execute(inputdata, mapper, reducer_unique)
```

Counting non unique friends...

```
["Myriel", 6]
["Napoleon", 1]
["MlleBaptistine", 4]
["MmeMagloire", 1]
["Champtercier", 1]
["Valjean", 16]
```

Counting unique friends...

```
["Myriel", 5]
["Napoleon", 1]
["MlleBaptistine", 3]
["MmeMagloire", 1]
["Champtercier", 1]
["Valjean", 16]
```

1.9 Problem 4: Aymmetric Followers

(6 points)

The relationship “friend” is often symmetric, meaning that if I am your friend, you are my friend. Implement a MapReduce algorithm to check whether this property holds. Generate a list of all non-symmetric friend relationships.

1.9.1 Map Input

Each input record is a two-element list [personA, personB] where personA is a string representing the name of a person and personB is a string representing the name of one of personA’s friends. Note that it may or may not be the case that the personA is a friend of personB.

1.9.2 Reduce Output

The output should be all pairs (friend, person) such that (person, friend) appears in the dataset but (friend, person) does not.

Correct output: asymmetric_friendships.json

```
[106]: # Part 0
filename = "friends.json"

# Part 1
mr = MapReduce()

# Part 2
def mapper(record):
    person, friend = record
    mr.emit_intermediate((person, friend), 1)

# Part 3
def reducer(key, list_of_values):
    # key: (person, friend)
    # Check if the reverse pair exists in the intermediate dict
    reverse_key = (key[1], key[0])
    if reverse_key not in mr.intermediate:
        mr.emit([key[1], key[0]])

# Part 4
with open(filename) as inputdata:
    asym_friends = mr.execute(inputdata, mapper, reducer)
# Check against answer file
answers = convert_hack_to_list('asymmetric_friendships.json')
for row in asym_friends:
    found = False
    for answer in answers:
        if row == answer:
            found = True
            break
    if not found:
        print(f"**Row {row} not found in answers.")
```

```
["Geborand", "Myriel"]
["Count", "Myriel"]
["OldMan", "Myriel"]
["Myriel", "Napoleon"]
["Myriel", "MlleBaptistine"]
["Valjean", "MlleBaptistine"]
["MmeMagloire", "MlleBaptistine"]
["Myriel", "MmeMagloire"]
["MmeMagloire", "Valjean"]
```



```

["Labarre", "Valjean"]
["Marguerite", "Valjean"]
["MmeDeR", "Valjean"]
["Isabeau", "Valjean"]
["Fantine", "Valjean"]
["Cosette", "Valjean"]
["Simplice", "Valjean"]
["Woman1", "Valjean"]
["Judge", "Valjean"]
["Woman2", "Valjean"]
["Gillenormand", "Valjean"]
["MlleGillenormand", "Valjean"]
["Babet", "Valjean"]
["Montparnasse", "Valjean"]

```

1.10 Problem 5: Genetic data

(6 points)

Consider a set of key-value pairs where each key is sequence id and each value is a string of nucleotides, e.g., GCTTCCGAAATGCTCGAA....

Design a MapReduce algorithm to remove the last 10 characters from each string of nucleotides, then remove any duplicates generated.

1.10.1 Map Input

Each input record is a two-element list (sequence id, nucleotides) where sequence id is a string representing a unique identifier for the sequence and nucleotides is a string representing a sequence of nucleotides

1.10.2 Reduce Output

The output from the reduce function should be the unique trimmed nucleotide strings.

Correct output: unique_trims.json

```

[107]: # Part 0
filename = "dna.json"

# Part 1
mr = MapReduce()

# Part 2
def mapper(record):
    mr.emit_intermediate(record[0], record[1][:-10]) # Trim the last 10
    ↪ characters from the nucleotide string

# Part 3
def reducer(key, list_of_values):

```

```

unique_nucleotides = set(list_of_values) # Use a set to remove duplicates
mr.emit((key, list(unique_nucleotides)))

# Part 4
with open(filename) as inputdata:
    dna_list = mr.execute(inputdata, mapper, reducer)
# Check against answer file
answers = convert_hack_to_list('unique_trims.json')
for row in dna_list:
    found = False
    for answer in answers:
        if row[1][0] == answer:
            found = True
            break
    if not found:
        print(f"***Row {row} not found in answers.")

```

```

["RATGRG", ["GGGGTGGCTACCCAGAGGCATGCTCCTCACCCAGCTCCACTGTCCCTACCTGCTGCTGCTGGTG
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```

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